Constraint Swap Reanalysis: A Marginal Signal at 32 Seeds

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Constraint-Swap Causal Geometry — intervention-algebra reanalysis **Status:** Overall **NO_GO** under preregistered gates. R1 GO, R2 GO, R3 NO_GO. Correlations between A-side and B-side intervention effects ARE below the 0.30 threshold (r_undo = -0.234, r_rescue = 0.079), consistent with the intervention-algebra prediction of near-independence. But permutation testing does not reject the "|r| ≥ 0.30" null at α = 0.05 (p_undo = 0.0965, p_rescue = 0.0972). With 32 seeds we are underpowered to distinguish "genuinely low correlation" from "not-far-enough-below-0.30 to reject." Third serial null-ish result on the intervention-algebra reframe, with a specific power-diagnostic that names what would be needed to decide it. **Date:** 2026-07-27

Abstract

The 32-seed Constraint-Swap Causal Geometry run rejected the constraint-specific-deformation claim on all five G1–G5 gates (REJECT_CONSTRAINT_SPECIFIC_DEFORMATION). The human director's intervention-algebra reframe proposed that the object may not live inside the representation but in the family of interventions under which behavior changes. Under that reframe, the univariate G3/G4 gates could miss constraint-specific structure that appears in the **joint distribution** of A-side and B-side intervention effects across seeds: if the effects are shared-artifact-driven they should be strongly correlated, if they are constraint-specific they should be approximately independent.

This reanalysis, preregistered before executing, tested three gates on the frozen 32-seed data:

- **R1** |r(undo_A_specific_harm, undo_B_specific_harm)| < 0.30
- **R2** |r(rescue_A_specific_gain, rescue_B_specific_gain)| < 0.30
- **R3** permutation test rejects |r| ≥ 0.30 at p < 0.05

Result: r_undo = **−0.234**, r_rescue = **0.079**. R1 and R2 both GO. R3 NO_GO (p_undo = 0.0965, p_rescue = 0.0972).

The correlations are **directionally consistent with the reframe** — low, and the negative sign on r_undo actually goes further in the independence direction than the reframe strictly requires. But the permutation test cannot reject the shared-artifact null at 32 seeds, because the sample is small enough that a true |r| between 0.20 and 0.30 is easily consistent with observed values around ±0.20. Under strict preregistration, R3 fails and the overall verdict is NO_GO.

Substantively: this is a *marginal signal*, not a clean null. The reframe's abstract prediction (low cross-condition correlation) is what the data shows. What the data cannot do is rule out that the same low correlation would appear under a shared-artifact null. A larger seed count — 100+ — could plausibly resolve it either way, because the observed r values are close enough to the threshold that sample-size doubling would shrink the confidence interval enough to matter.

Third serial preregistered null on the intervention-algebra reframe, with a specific quantitative note about what would decide it.

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1. What was preregistered

REANALYSIS_INTERVENTION_ALGEBRA_PREREGISTRATION.md (2026-07-27, before reanalysis_intervention_algebra.py was drafted or executed):

results/registered_seed_rows.jsonl. No new training, no new intervention data.

metrics G3/G4 tested univariately): - undo_A_specific_harm - undo_B_specific_harm - rescue_A_specific_gain - rescue_B_specific_gain

(permutation test at $\alpha = 0.05$ rejects $|r| \geq 0.30$).

correlations are low; if shared artifact dominates, correlations are high.

- Data source: frozen 32 seed rows from
- Four preregistered intervention effects (from the primary topology
- Three gates: R1 ($|r_{undo}| < 0.30$), R2 ($|r_{rescue}| < 0.30$), R3
- Prediction: if the intervention-algebra reframe is right,

2. Results

Pearson correlations across 32 seeds:

- r(undo_A_specific_harm, undo_B_specific_harm) = **−0.234**
- r(rescue_A_specific_gain, rescue_B_specific_gain) = **0.079**

Permutation test (10,000 label-shuffles at seed 20260727):

- p(|r_undo| ≥ 0.30 under H0) = **0.0965**
- p(|r_rescue| ≥ 0.30 under H0) = **0.0972**

Descriptive statistics of the four intervention effects (across-seed):

effect	mean	std	min	max
undo_A_specific_harm	−0.225	0.197	−0.599	0.000
undo_B_specific_harm	−0.187	0.119	−0.513	0.000
rescue_A_specific_gain	~0.06	~0.14	−0.20	~0.30
rescue_B_specific_gain	−0.057	0.140	−0.276	0.372

3. Gate decisions

gate				
R1	r_undo	< 0.30	GO	0.234
R2	r_rescue	< 0.30	GO	0.079
R3 permutation rejects	NO_GO	p_undo 0.0965, p_rescue 0.0972		

Overall NO_GO.

Licensed reading (from the runner):

> *correlations below 0.30 but permutation test does not reject*

The strict-preregistration verdict is NO_GO. The substantive picture is that the observed correlations are *near* what the reframe predicts (well below 0.30) but the sample size is small enough that the permutation test cannot distinguish "genuinely near zero" from "near 0.30 but not quite there."

4. What this actually tells us

Two things worth naming:

(1) The reframe's directional prediction is empirically consistent with the data. r_undo = −0.234 and r_rescue = 0.079 are both well below 0.30 in absolute value. A shared-artifact null (where all seeds share a common noise direction that flips A-side and B-side effects similarly) would produce |r| ≥ 0.5 easily. That is not what we see.

(2) The evidence is not strong enough to rule out a moderate-r shared-artifact null. With 32 seeds, a true correlation of ±0.25 produces observed |r| ≥ 0.30 with substantial probability. That's what R3's failure captures. To decisively reject "the effects are 0.30-correlated," we'd need approximately 100 seeds — not just to tighten the estimate but because the R3 permutation test is comparing observed to threshold, not observed to zero.

If someone were to point at this result and say "*the reframe survived*", they'd be over-reaching. If someone were to say "*the reframe was killed*", they'd also be over-reaching. What actually happened: the preregistered null-hypothesis rejection didn't clear, and the data are directionally consistent with the reframe but the sample is small.

This is exactly the outcome where preregistration discipline matters most. A post-hoc reader could tell either story from the data. The preregistered decision rule tells one specific story: NO_GO. Honor it.

5. What would decide it

0.20 and R3 rejects, reframe wins. If |r| drifts up toward 0.30 and R3 stays unrejected or flips positive, shared-artifact dominates.

are rank-4 affine transports pre-specified from theory. An intervention sweep (many random transports per seed, cluster by behavioral effect) might reveal constraint-specific structure the four preregistered directions miss. This is the DR6-style sweep the original reframe suggested and this reanalysis could not do with the frozen data.

- **~100 more seeds** with the same protocol. If |r| stays near
- **A different intervention design.** The current interventions

6. What the reanalysis licenses

narrow test of the correlation-structure prediction on the four preregistered intervention effects.

Constraint Swap with the same protocol would decide R3 one way or the other. That is a real next experiment.

can extract additional signal, but their power depends on the original sample size. 32 seeds was appropriate for the original G1–G5 gates but is underpowered for correlation-based reanalyses.

- **Not a full test of the intervention-algebra reframe.** Only a
- **A specific follow-up justification.** A 100-seed replication of
- **A methodological note.** Reanalyses of preregistered data

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Appendix: reproduction

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uv run --no-sync python -m experiments.constraint_swap_causal_geometry.reanalysis_intervention_algebra
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Reads the frozen results/registered_seed_rows.jsonl, computes Pearson correlations and permutation p-values (10,000 trials, seed 20260727), applies R1/R2/R3 gates, writes results/reanalysis_intervention_algebra.json. Local CPU, seconds.

Preregistration digest (SHA-256 of REANALYSIS_INTERVENTION_ALGEBRA_PREREGISTRATION.md**):** 4715fd76531a9977ccd3cef3fe685f9567ba0517068e0b50646b69ceccf5c1ed.